

SHENKURSK, Russian Federation

WMO: 227680

Lat: 62.1140N

Lon: 42.8899E

Elev: 45

StdP: 100.79

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.1	(c) -28.7	(d) -34.9	(e) 0.1	(f) -32.1	(g) -31.6	(h) 0.2	(i) -28.8	(j) 8.4	(k) -6.2	(l) 7.7	(m) -5.9	(n) 1.0	(o) 190	(p) 0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.2	(c) 28.7	(d) 19.0	(e) 26.6	(f) 18.1	(g) 24.6	(h) 17.1	(i) 20.4	(j) 26.2	(k) 19.3	(l) 24.7	(m) 18.2	(n) 23.1	(o) 3.1	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	13.3	22.9	17.3	12.4	21.8	16.2	11.6	20.8	58.9	26.1	55.1	24.9	51.7	23.2	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.9	(b) 7.0	(c) 6.2	(e) -36.1	(f) 31.1	(g) 3.6	(h) 1.8	(i) -38.7	(j) 32.5	(k) -40.8	(l) 33.5	(m) -42.8	(n) 34.6	(o) -45.3	(p) 35.9		
(a) 7.9	(b) 7.0	(c) 6.2	(e) -36.2	(f) 21.8	(g) 3.5	(h) 1.4	(i) -38.8	(j) 22.8	(k) -40.8	(l) 23.6	(m) -42.8	(n) 24.4	(o) -45.4	(p) 25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-12.1	-11.2	-5.2	2.7	9.3	14.5	18.0	14.8	9.6	3.2	-4.3	-8.8		
	DBStd	11.88	8.86	8.45	5.66	4.96	5.33	4.59	4.11	3.99	3.80	4.14	6.53	8.13		
	HDD10.0	3349	686	594	470	223	80	10	1	4	53	215	431	582		
	HDD18.3	5819	944	827	728	468	282	132	58	125	263	471	681	841		
	CDD10.0	656	0	0	0	6	59	146	249	152	40	3	0	0		
	CDD18.3	83	0	0	0	0	3	18	48	14	0	0	0	0		
	CDH23.3	709	0	0	0	1	31	177	392	108	1	0	0	0		
	CDH26.7	173	0	0	0	0	4	47	94	27	0	0	0	0		
(14) Wind	WSAvg	3.0	3.1	3.1	3.2	3.0	3.1	2.9	2.6	2.6	2.8	3.2	3.1	3.3		
(15) Precipitation	PrecAvg	568	33	27	28	37	50	64	69	77	54	50	40	40		
	PrecMax	747	67	50	81	117	86	155	160	180	112	94	72	94		
	PrecMin	386	8	0	4	5	15	14	21	13	16	1	12	15		
	PrecStd	76	12	14	17	22	22	32	35	37	22	18	14	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	2.6	8.4	20.6	27.0	30.7	31.5	29.9	22.4	15.6	7.7	4.5		
		MCWB	1.4	1.5	3.1	12.1	15.5	19.3	20.7	20.4	15.8	11.5	6.4	3.6		
	2%	DB	1.2	1.5	5.0	16.5	23.5	27.8	29.1	26.3	19.2	12.4	5.4	2.4		
		MCWB	0.5	0.6	1.5	9.3	13.9	18.1	19.6	18.5	14.5	10.0	4.3	1.5		
	5%	DB	0.4	0.6	3.1	12.5	21.0	25.1	27.3	23.5	17.1	10.5	3.9	1.4		
		MCWB	-0.2	-0.1	0.8	6.9	12.5	16.7	18.9	17.2	13.4	8.8	3.3	0.7		
	10%	DB	-1.0	-0.5	2.0	10.0	18.4	22.8	25.5	21.4	15.3	9.0	2.5	0.6		
		MCWB	-1.6	-1.2	0.4	5.3	11.3	15.5	18.1	16.2	12.5	7.7	1.7	0.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	1.7	3.9	12.5	17.4	20.8	22.3	21.5	16.5	12.3	6.6	3.6		
		MCDB	2.0	2.4	7.3	20.2	23.5	27.9	28.3	27.6	20.0	14.4	7.4	4.1		
	2%	WB	0.6	0.7	2.3	9.8	15.2	19.0	20.8	19.5	15.1	10.5	4.5	1.7		
		MCDB	1.1	1.3	4.0	14.9	21.1	25.9	26.9	24.3	18.1	12.0	5.3	2.2		
	5%	WB	-0.1	0.0	1.4	7.6	13.7	17.8	19.8	18.0	13.9	9.0	3.3	0.7		
		MCDB	0.5	0.7	2.7	12.0	18.9	23.2	25.2	22.4	16.5	10.3	3.9	1.2		
	10%	WB	-1.6	-1.2	0.6	5.6	12.2	16.5	19.0	16.8	12.8	7.6	1.8	0.0		
		MCDB	-1.1	-0.5	1.9	9.4	17.6	21.4	24.2	20.3	15.2	8.9	2.3	0.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.9	8.3	8.5	9.3	9.4	9.2	8.2	6.7	4.2	4.0	5.1		
		MCDBR	5.3	4.7	8.0	12.5	13.2	12.7	12.1	11.2	9.7	6.5	4.0	3.6		
	5% WB	MCWBR	5.1	4.4	5.7	6.9	5.8	5.3	4.6	4.7	5.5	4.5	3.7	3.6		
		MCWBR	5.0	4.7	5.9	11.6	10.9	11.1	10.7	9.8	8.4	6.0	3.9	3.8		
(40) Clear-Sky Solar Irradiance	taub		0.246	0.297	0.325	0.364	0.354	0.353	0.391	0.371	0.338	0.317	0.266	0.220		
	taud		1.999	2.065	2.093	2.198	2.438	2.469	2.373	2.404	2.484	2.377	2.107	1.956		
	Ebn at Noon		407	633	764	816	866	872	822	809	773	636	385	249		
	Edh at Noon		38	74	103	115	99	98	105	94	72	55	34	21		
(44) All-Sky Solar Radiation	RadAvg		0.24	0.90	2.26	3.81	4.89	5.24	5.19	3.63	2.05	0.78	0.23	0.11		
	RadStd		0.03	0.20	0.36	0.39	0.32	0.57	0.43	0.38	0.24	0.11	0.05	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		(d) +0.70	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47) Regional (3 neighbors)		(d) +0.72	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) -222	(k) -253	(l) N/A	(m) N/A

Nomenclature: See separate page